SUMMARY

Let S be the sample space associated with a random experiment and A be an event.

$$P(A) = \frac{n(A)}{n(S)} = \frac{\text{Number of cases favourable to } A}{\text{Exhaustive Number of cases in } S}$$

Axioms of probability

Given a finite sample space S and an event A in S, we define $P(A)$, the probability of A, satisfies the following three axioms.

(1) $P(A) \geq 0$

(2) If A and B are mutually exclusive events, then $P(A \cup B) = P(A) + P(B)$

(3) $P(S) = 1$

The probability of the impossible event is zero. That is $\boxed{P(\varnothing) = 0}$

If A and B are any two events and $\overline{B}$ is the complementary events of B, then

$$\boxed{P(A \cap \overline{B}) = P(A) - P(A \cap B)}$$

If A and B are any two events, then

$$\boxed{P(A \cup B) = P(A) + P(B) - P(A \cap B)}$$

The conditional probability of an event B, assuming that the event A has already happened is denoted by $P(B/A)$ and is defined as

$$\boxed{P(B/A) = \frac{P(A \cap B)}{P(A)} \quad \text{provided } P(A) \neq 0}$$

$$\boxed{P(A/B) = \frac{P(A \cap B)}{P(B)} \quad \text{provided } P(B) \neq 0}$$

The probability of the simultaneous happening of two events A and B is given by

$$\boxed{P(A \cap B) = P(A/B)P(B) \text{ or } P(A \cap B) = P(B/A)P(A)}$$

Two events A and B are said to be **independent** if and only if

$$\boxed{P(A \cap B) = P(A) \,.\, P(B)}$$

Total Probability

If $A_1, A_2, A_3, \ldots A_n$ are mutually exclusive and exhaustive events and B is any event in S then $P(B)$ is called the total probability of event B and

$$\boxed{P(B) = P(A_1).P(B/A_1) + P(A_2).P(B/A_2) + \cdots + P(A_n).P(B/A_n) = \sum_{i=1}^{n} P(A_i).P(B/A_i)}$$

Bayes' Theorem

If $A_1, A_2, A_3, \ldots A_n$ are mutually exclusive and exhaustive events such that $P(A_i) > 0$, $i = 1, 2, 3, \ldots . n$ and B is any event in with $P(B) > 0$, then

$$\boxed{P(A_i/B) = \frac{P(A_i)\,P(B/A_i)}{P(A_1)\,P(B/A_1) + P(A_2)\,P(B/A_2) + \ldots + P(A_n)\,P(B/A_n)}}$$

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